

Ruthvek Kannan

[LinkedIn](#) / [Github](#) / [Portfolio](#)

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TECHNICAL SKILLS

- **Languages:** Python, C, TypeScript/JavaScript, SQL, HTML/CSS
- **Backend & Infrastructure:** Node.js, FastAPI, REST APIs, Celery, PostgreSQL, pgvector, Redis, Docker, Kubernetes, OpenTelemetry
- **Applied AI:** LangGraph, RAG (Hybrid Search, RRF), LiteLLM, Multi-Agent Orchestration
- **Frontend:** React, Next.js, Tailwind CSS
- **Cloud & DevOps:** AWS, Vercel, Render, Supabase, Git, GitHub Actions (CI/CD)

EDUCATION

- **Symbiosis Institute of Technology, Nagpur (Symbiosis International Deemed University)** Nagpur, India
Bachelor of Technology in Computer Science & Engineering with Honours in Data Science 2022 – 2026

EXPERIENCE

- **Apoliums Infotech India Pvt. Ltd.** India
Software Engineer Intern Dec 2025 – Jun 2026
 - Developed and shipped full-stack web and mobile applications across multiple client domains using JavaScript/TypeScript, optimizing system performance and responsiveness.
 - Designed and implemented robust REST APIs and scalable backend architectures, seamlessly integrating third-party AI provider capabilities as auxiliary features.
 - Collaborated with cross-functional engineering teams to build resilient cloud services, enforce CI/CD pipelines, and ensure high availability of production systems.

PROJECTS

- **AEGIS – Distributed Code Verification System** [Live Demo](#) | [GitHub](#)
Python, FastAPI, LangGraph, React, Docker, PostgreSQL, pgvector
 - Architected a distributed event-driven verification loop utilizing applied AI models to enforce a Red-Green-Refactor CI/CD pipeline, requiring proof of a failing test before any fix is merged.
 - Built isolated, non-root Docker sandboxing for all LLM-generated code execution, with a graceful local-subprocess fallback ensuring uptime independent of container infrastructure.
- **Athenis AI – Scalable Enterprise Search & RBAC Platform** [Live Demo](#) | [GitHub](#)
Python, FastAPI, Next.js, Celery, PostgreSQL, Redis, Kubernetes
 - Engineered a high-performance search backend combining PostgreSQL full-text search (BM25) with vector similarity, utilizing Celery asynchronous queues for document processing.
 - Designed an extensible abstraction layer supporting multiple third-party API providers, ensuring seamless integration and high availability with zero code changes during provider swaps.
- **Project Watchdog – Event-Driven Data Processing Pipeline** Academic Capstone
Python, Gemini API, Whisper, Streamlit
 - Designed and developed an autonomous file-monitoring daemon that detects new datasets in real-time, leveraging event-driven triggers to execute downstream processing tasks.
 - Automated exploratory data analysis: LLM-generated visualization code and AI-written interpretation are executed and compiled into a multi-page PDF report via fpdf2.

ACHIEVEMENTS

- **Indian Patent Published: Machine learning based interactive dashboard system**
Publication No: IN202621032990 A1 · Interactive dashboard for crop cost prediction and profitability analysis using ML algorithms.
- **IEEE Research Paper: Guidelines for Effective UI/UX Design on Cancer Support NGO Websites**
Published in 2025 International Conference on AI and Machine Vision (AIMV) · Oct 2025
- **IEEE Research Paper: Architectural Innovations for AI Chips, Testing, and High Performance Computing**
Published in 2024 IEEE 6th Intl Conf on Cybernetics, Cognition and ML Applications (ICCCMLA) · Feb 2025

CERTIFICATIONS

AWS Cloud Foundations, AWS Cloud Security, Oracle Database Foundations, Red Hat System Administration I, ServiceNow CSA, ServiceNow CAD, Infosys Big Data.